Blessed G/Michael Catholic Secondary School

Ist Semester mathematics test one for Grade 9th

oct , 2018 E.C

Time Allowed :- —1;30hr

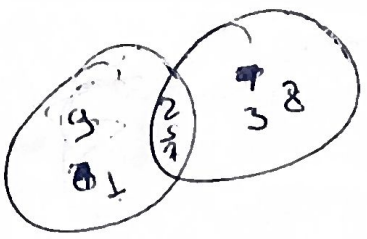
Write true if the statement is correct false otherwise.

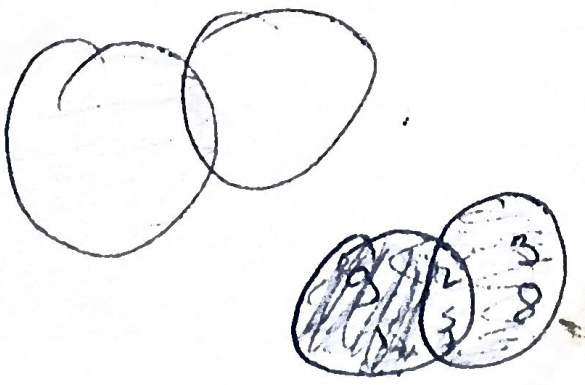
1. A set is a collection of well-defined objects or elements .
2. A set which does not contain any element is called an void set .
3. A set which consists of a definite number of elements is called a finite set.
4. Two sets A and B are said to be equal if and only if they have exactly the same or identical elements.
5. If $A \subseteq B$ and $A \neq B$, then A is called the proper subset of set B.
6. Two sets A and B are disjoint if $A \cap B = \emptyset$.
7. x is said to be in simplest (lowest form) if a and b are relatively prime or GCF(a, b) = 1
8. Every natural number can be expressed (factored) as a product of primes and this factorization is unique.
9. If a is rational number and b irrational number then ab is rational number.
10. For two positive integers a and b in the division lemma, we say a is divisible by b if the remainder r is zero.
11. If a is a natural number then it is either prime or composite.

Given U={1,2,3,4,5,6,7,8,9} A={2,5,9,7,1} B={7,2,3,5,8} based on the given sets

Match column A with column B

A	B
12. A′ n B′	A.{ }
13. A Δ B	B.{4,6}
14. (A)B U A	C.{1,3,8,9}
15. A\B′	D.{2,5,7}





16. Find GCF (12, 18) × LCM(12, 18).

A. 216 B. 200 C. 36 D.108

17. In a book store there are 125 different books to be distributed to 9 students. If the book store shares these books equally, how many books will each student receive and how many books will be left?

A. 8 &13 B. 9&13 C.8&9 D.13&8

18. Which of the following are well defined sets

A. Collection of students in your class.

B. Collection of beautiful girls in your class.

C. Collection of consonants of the English alphabet.

D. The collection of kind students in your school

E. a & c

19. Suppose that A is the set of positive even numbers. Which of the following is true

A. $4 \in A$ B. $5 \in A$ C. $-2 \in A$ D. $0 \notin A$

20. Which f the following is infinite set

A. $A = \{x \mid x \in$ and $5 < x < 6\}$

B. $B = \{0\}$

C. C is the set of odd natural numbers divisible by 2.

D. $D = \{\}$

21. Which of the following is false

A . Any set is a subset of itself.

B . Empty set is a subset of every set.

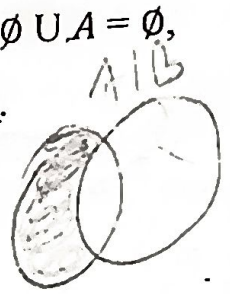
C .For any set A, A is not a proper subset of itself.

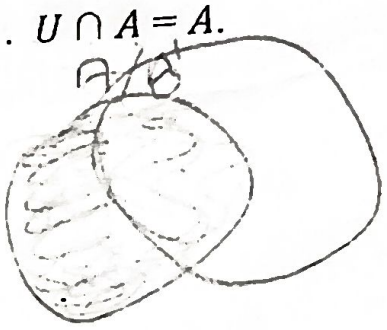
D . The number of proper subsets of set A is $2n - 1$.

E.Empty set is the proper subset of any other sets.

22. Which of the following is false

A. $(A \cup B)' = A' \cap B'$ B . $(A \cap B)' = A' \cup B'$ C. $\emptyset \cup A = \emptyset$, D . $U \cap A = A$.





23. For any two sets A and B, which of the following is false.

A. $(A')' = A$

B. $A \subseteq B \Leftrightarrow B' \subseteq A$

C. $A - B = A \cap B'$

D. $A \Delta B = (A \backslash B) \cup (B \backslash A)$

24. Given $A = \{0, 2, 3, 4, 5\}$ and $B = \{0, 1, 2, 3, 5, 7, 9\}$. Then, find $A \Delta B$.

A.{0,1,3} B.{4,7,9} C. empty set D.{0,2,3,5}

25. If the number of elements of set A is 7 and the number of elements of set B is 4, then the number of elements of $A \times B$ is _____

A. 24 B. 25 C.28 D. 11

26. For any two sets A and B, which of the following is false.

A. $n(A \cup B) = n(A) + (B) - n(A \cap B)$

B. $n(A') = n(U) - n(A)$

C. $n(A \backslash B) = n(A) - n(B)$

D. $n(A \Delta B) = n(A) + n(B) - 2n(A \cap B)$

27. Consider a universal set $U = \{1, 2, 3 \ldots 14\}$, $A = \{1, 3, 5, 7, 11\}$, $B = \{2, 4, 8, 9, 10, 11\}$.
Then which one of the following is true?

A. $(A \cup B)' = \{1, 4, 6, 12, 13, 14\}$

B. $A \cap B = A' \cup B'$

C. $A \Delta B = (A \cap B)'$

D. $A \setminus B = \{3, 5, 7\}$

Short answer

28 .In a survey of 200 students in Motta Secondary School, 80 students are members of Nature club, 30 students are members of Mini-media club, 20 students are members of both clubs. Answer the following questions.

a. How many students are members of either of the clubs?

b. How many students are not members of either of the clubs?

c. How many students are only in Nature club?

29. A survey was conducted in a class of 100 children and it was found out that 45 of them like only Mathematics whereas 35 like Science and 20 students like both subjects. How many like neither of the subjects?

30. Find the remainder of $\frac{(5(m+1))(5m+3)(5m+6)}{5}$ for m is non-negative integer

31. .Let set $B = \{0, 2, 4, \ldots\}$. Described this set in builder method

32. If $A \Delta B = \emptyset$, then what can be said about the two sets?

33. If $A = \{0, 1, 2\}$, then find $A \times A$.

34. If the perimeter of a triangle is 12 and lengths of the sides are natural numbers, find all the possible lengths of sides of the triangle

Workout(show all necessary steps)

35. Find the greatest common factors (GCF) of 180, 270, 1,080 (using prim factorization)

36. Consider two natural numbers 15 and 42. Then, find LCM(15,42) (using prim factorization)

37. Express $-3\frac{3}{5}$ as improper fraction.

38 .Given that 77,9 $x2$ is a number where x is its tens place. If this number is divisible by 6, what is (are)the possible value(s) of x?

Name: Mulatu Solomon Section: D No: 40

1. True
2. True
3. True
4. True
5. False
6. True
7. True
8. True
9. False
10. True
11. False
12. B
13. C
14. A
15. D
16. A
17. D
18. E
19. A
20. A
21. D
22. C
23. C
24. B
25. C
26. C
27. C D

28. A. ~~30 60 students~~ ~~100 students~~ ~~50 students 60~~ B. C.
29. 40
30. 0
31. $\{2x \mid x \in \mathbb{W}\}$ or $\{x \mid x \in \mathbb{W}, x \text{ is divisible by } 2\}$
32. The two sets contain the same element
33. $\{0, 0.5, 1, 1.5, 2, 2.5, 3, 3.5, \{2,2\}, \{2,2\}\}$
34. $(1,1), (2,2)$ (4,4) (3,3) (5,5)

WORKOUT PART

35. 1st Prime factorization.
$180 = 2^2 \times 3^2 \times 5$
$270 = 3^3 * 2 * 5$
$1080 = 2^3 * 3^3 * 5$
2nd take the common factor with lowest exponent
$2, 3^2, 5$
3rd take the Product $= 90$

(x, y) - Order pair
{x, y} - Set

36. 1st Prime factorization
$15 = 3 \times 5$ {2, 3, 5}
$42 = 3 \times 2 \times 7$
2nd take common factor with high exponent.
→ Therefore 15 & 42 are relatively Prime nos b/c they have no common factor.
~~GCF(15,42) = 3 15x~~ $LCM(15, 42) = \frac{15 \times 42}{5} = \frac{630}{5}$

37. $-3\frac{3}{5} = \frac{(-3 \times 5) + 3}{5} = \frac{-12}{5} = -\frac{18}{5}$
correction $-\frac{(3 * 5 + 3)}{5} = -\frac{15}{5}$

38. 2, 5, Het c